

The Chain Rule & Directional Derivatives

ANSWER KEY



The multivariable chain rule

- 1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
 $\frac{\partial z}{\partial x} = 2x$, $\frac{\partial z}{\partial y} = 2y$, $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$. $\frac{dz}{dt} = 2x(-\sin t) + 2y \cos t = -2 \cos t \sin t + 2 \sin t \cos t = 0$
(consistent with $z = \cos^2 t + \sin^2 t = 1$, a constant).
- 2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
 $\frac{\partial w}{\partial x} = y$, $\frac{\partial w}{\partial y} = x$, $\frac{\partial w}{\partial z} = 2z$. $\frac{dw}{dt} = y(1) + x(2t) + 2z(3t^2)$. At $t = 1$: $x = y = z = 1$, so $\frac{dw}{dt} = 1 + 2 + 6 = 9$.
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Chain rule with two independent variables

- 3 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 $\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y}$.
- 4 For $z = x^2 y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2$, $t = 1$.
At $s = 2$, $t = 1$: $x = 3$, $y = 1$. $\frac{\partial z}{\partial x} = 2xy = 6$, $\frac{\partial z}{\partial y} = x^2 = 9$. Since $\frac{\partial x}{\partial s} = 1$, $\frac{\partial y}{\partial s} = 1$: $\frac{\partial z}{\partial s} = 6 + 9 = 15$. Since $\frac{\partial x}{\partial t} = 1$, $\frac{\partial y}{\partial t} = -1$: $\frac{\partial z}{\partial t} = 6 - 9 = -3$.
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Gradient and directional derivative

- 5 Find the gradient of $f(x, y) = x^2 y^3$ at $(1, 2)$.
 $\nabla f = \langle 2xy^3, 3x^2 y^2 \rangle$. At $(1, 2)$: $\langle 2(8), 3(4) \rangle = \langle 16, 12 \rangle$.
- 6 Find the directional derivative of $f(x, y) = x^2 y$ at $(2, 1)$ in the direction of $\mathbf{v} = \langle 1, 1 \rangle$.
 $\nabla f = \langle 2xy, x^2 \rangle$; at $(2, 1)$: $\langle 4, 4 \rangle$. Unit vector $\hat{\mathbf{v}} = \langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$. $D_{\mathbf{v}} f = \frac{4}{\sqrt{2}} + \frac{4}{\sqrt{2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2}$.
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The gradient points in the direction of steepest ascent

- 7 For $f(x, y) = 3x^2 - y^2$ at $(1, 2)$, find the direction of steepest ascent and the maximum rate of increase.
 $\nabla f = \langle 6x, -2y \rangle$; at $(1, 2)$: $\langle 6, -4 \rangle$. Direction of steepest ascent: $\langle 6, -4 \rangle$. Maximum rate of increase:
 $|\nabla f| = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$.
- 8 In what direction from $(1, 1)$ does $f(x, y) = xy$ decrease most rapidly, and at what rate?
 $\nabla f = \langle y, x \rangle$; at $(1, 1)$: $\langle 1, 1 \rangle$. Steepest decrease is $-\nabla f = \langle -1, -1 \rangle$, at rate $|\nabla f| = \sqrt{2}$.
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Application of the gradient

- 9 The temperature on a plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at $(2, 1)$ wants to cool off fastest. In which direction should it move?
 $\nabla T = \langle -2x, -4y \rangle$; at $(2, 1)$: $\langle -4, -4 \rangle$ (direction of steepest warming). To cool fastest, move in $\langle 4, 4 \rangle$ (or the unit direction $\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \rangle$).
- 10 Find the rate of change of $T(x, y) = 100 - x^2 - 2y^2$ at $(2, 1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.
 $\nabla T(2, 1) = \langle -4, -4 \rangle$ (from above). $D_{\mathbf{v}}T = \langle -4, -4 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = -\frac{12}{5} - \frac{16}{5} = -\frac{28}{5}$.

More chain-rule practice

- 11 Let $z = x^2 - y^2$, $x = e^t$, $y = e^{-t}$. Find $\frac{dz}{dt}$.
 $\frac{\partial z}{\partial x} = 2x$, $\frac{\partial z}{\partial y} = -2y$; $\frac{dx}{dt} = e^t$, $\frac{dy}{dt} = -e^{-t}$. $\frac{dz}{dt} = 2xe^t + 2ye^{-t}$. Substituting $x = e^t$, $y = e^{-t}$: $\frac{dz}{dt} = 2e^{2t} + 2e^{-2t}$.
- 12 Let $w = x^2 + y^2 + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
 $\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = 2z$. $\frac{dw}{dt} = 2x(1) + 2y(2t) + 2z(3t^2)$. At $t = 1$: $x = y = z = 1$, so $\frac{dw}{dt} = 2 + 4 + 6 = 12$.

More gradient and directional derivative practice

- 13 Find the gradient of $f(x, y) = e^{xy}$ at $(1, 0)$.
 $\nabla f = \langle ye^{xy}, xe^{xy} \rangle$. At $(1, 0)$: $\langle 0, 1 \rangle$.
- 14 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(3, 4)$ in the direction of $\mathbf{v} = \langle -4, 3 \rangle$.
 $\nabla f = \langle 2x, 2y \rangle$; at $(3, 4)$: $\langle 6, 8 \rangle$. Unit vector: $|\mathbf{v}| = 5$, so $\hat{\mathbf{v}} = \langle -\frac{4}{5}, \frac{3}{5} \rangle$. $D_{\mathbf{v}}f = 6(-\frac{4}{5}) + 8(\frac{3}{5}) = -\frac{24}{5} + \frac{24}{5} = 0$
 (this direction is tangent to the level curve through $(3, 4)$).

The tangent line to a level curve

- 15 Using that ∇f is normal to the level curve of f , find the tangent line to $x^2 + y^2 = 25$ at $(3, 4)$.
 For $f(x, y) = x^2 + y^2$, $\nabla f = \langle 2x, 2y \rangle$; at $(3, 4)$: $\langle 6, 8 \rangle$, normal to the tangent line. Tangent line:
 $6(x - 3) + 8(y - 4) = 0 \Rightarrow 6x + 8y = 50 \Rightarrow 3x + 4y = 25$.

The chain rule with three intermediate variables

- 16 For $w = f(x, y, z)$ with $x = g(s, t)$, $y = h(s, t)$, $z = k(s, t)$, write the chain rule formula for $\frac{\partial w}{\partial t}$.
 $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial t}$.

One more gradient computation

- 17 Find the gradient of $f(x, y) = \ln(x^2 + y^2)$ at $(1, 2)$.
 $\nabla f = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle$. At $(1, 2)$: $\left\langle \frac{2}{5}, \frac{4}{5} \right\rangle$.

- 18** Is the direction $\langle 1, -1 \rangle$ a direction of increase or decrease for $f(x, y) = x + y$ at any point?
 $\nabla f = \langle 1, 1 \rangle$ everywhere. $\nabla f \cdot \langle 1, -1 \rangle = 1 - 1 = 0$, so this direction is tangent to the level curves — neither increase nor decrease.
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A closing chain-rule problem

- 19** Let $z = \sin(x) \cos(y)$, $x = t^2$, $y = t$. Find $\frac{dz}{dt}$ at $t = 0$.
 $\frac{\partial z}{\partial x} = \cos x \cos y$, $\frac{\partial z}{\partial y} = -\sin x \sin y$; $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 1$. $\frac{dz}{dt} = 2t \cos x \cos y - \sin x \sin y$. At $t = 0$: $x = 0$, $y = 0$,
so $\frac{dz}{dt} = 0(\cos 0 \cos 0) - \sin 0 \sin 0 = 0$.
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A final directional derivative problem

- 20** Find the directional derivative of $f(x, y) = x^2 - y^2$ at $(2, -1)$ in the direction from $(2, -1)$ toward $(5, 3)$.
Direction vector: $\langle 5 - 2, 3 - (-1) \rangle = \langle 3, 4 \rangle$, $|\langle 3, 4 \rangle| = 5$, so $\hat{\mathbf{v}} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$. $\nabla f = \langle 2x, -2y \rangle$; at $(2, -1)$: $\langle 4, 2 \rangle$.
 $D_{\hat{\mathbf{v}}} f = 4 \left(\frac{3}{5} \right) + 2 \left(\frac{4}{5} \right) = \frac{12}{5} + \frac{8}{5} = 4$.
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Implicit differentiation via the chain rule

- 21** If $x^2y + y^3 = 8$ defines y implicitly as a function of x , use $\frac{dy}{dx} = -\frac{F_x}{F_y}$ (for $F(x, y) = x^2y + y^3 - 8$) to find $\frac{dy}{dx}$.
 $F_x = 2xy$, $F_y = x^2 + 3y^2$. So $\frac{dy}{dx} = -\frac{2xy}{x^2 + 3y^2}$.
- 22** Evaluate $\frac{dy}{dx}$ from the previous question at the point $(0, 2)$ (check first: $0^2(2) + 2^3 = 8$ ✓, so this point is on the curve).
 $\frac{dy}{dx} = -\frac{2xy}{x^2 + 3y^2}$. At $(0, 2)$: $-\frac{2(0)(2)}{0 + 3(4)} = -\frac{0}{12} = 0$.
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A last pair of gradient problems

- 23** Find the gradient of $f(x, y, z) = x^2 + y^2 + z^2$ at $(1, 2, 2)$, and find its magnitude.
 $\nabla f = \langle 2x, 2y, 2z \rangle$; at $(1, 2, 2)$: $\langle 2, 4, 4 \rangle$. Magnitude: $\sqrt{4 + 16 + 16} = \sqrt{36} = 6$.
- 24** For $f(x, y) = x^2y$, find the maximum rate of change of f at $(3, 1)$ and the direction in which it occurs.
 $\nabla f = \langle 2xy, x^2 \rangle$; at $(3, 1)$: $\langle 6, 9 \rangle$. Maximum rate of change: $|\nabla f| = \sqrt{36 + 81} = \sqrt{117} = 3\sqrt{13}$, occurring in the direction $\langle 6, 9 \rangle$.

Chain rule with a three-step composition

- 25** Let $z = x^2y$, $x = r \cos \theta$, $y = r \sin \theta$ (polar substitution). Find $\frac{\partial z}{\partial r}$.

$$\frac{\partial z}{\partial x} = 2xy, \frac{\partial z}{\partial y} = x^2; \frac{\partial x}{\partial r} = \cos \theta, \frac{\partial y}{\partial r} = \sin \theta. \frac{\partial z}{\partial r} = 2xy \cos \theta + x^2 \sin \theta.$$

- 26** Using the substitution above, evaluate $\frac{\partial z}{\partial r}$ at $r = 2$, $\theta = 0$.

$$\text{At } \theta = 0: x = r \cos 0 = r = 2, y = r \sin 0 = 0. \frac{\partial z}{\partial r} = 2(2)(0) \cos 0 + (2)^2 \sin 0 = 0 + 0 = 0.$$

A last directional-derivative setup

- 27** Find the directional derivative of $f(x, y) = xe^y$ at $(1, 0)$ in the direction $\theta = \frac{\pi}{3}$ from the positive x -axis.

$$\hat{\mathbf{v}} = \langle \cos \frac{\pi}{3}, \sin \frac{\pi}{3} \rangle = \left\langle \frac{1}{2}, \frac{\sqrt{3}}{2} \right\rangle. \nabla f = \langle e^y, xe^y \rangle; \text{ at } (1, 0): \langle 1, 1 \rangle. D_{\hat{\mathbf{v}}}f = 1 \left(\frac{1}{2} \right) + 1 \left(\frac{\sqrt{3}}{2} \right) = \frac{1 + \sqrt{3}}{2}.$$